

LECTURE - 9

Master Theorem:

$$T(n) = a T(n/b) + f(n)$$

$D(n)$ $C(n)$

Water shed func = $n^{\log_b(a)}$

Case 1: If there exist a constant $\epsilon > 0$ such that
 $f(n) = O(n^{\log_b(a) - \epsilon})$ ^{watershed} this provides upper bound to $f(n)$
 $\therefore T(n) = O(n^{\log_b(a)})$

Case 2: If there exist a constant $k > 0$ such that
 $f(n) = O(n^{\log_b(a)} \cdot \log^k(n)) \rightarrow$ growth of $f(n)$ & watershed is almost same as we don't

$\therefore T(n) = O(n^{\log_b(a)} \cdot \log^{k+1}(n))$ consider logarithmic growth that much - we consider exponential

k will be max

power of log func. in $f(n)$

Case 3: if there exist a constant $\epsilon \geq 0$ such that
 $f(n) = \Omega(n^{\log_b(a) + \epsilon})$ watershed provides lower bound to $f(n)$

Check for regularity constraint: $a f(n/b) \leq c \cdot f(n)$
 this should satisfy for case 3 (for constant $c < 1$).

Case 1: WD increases down the tree

Case 2: WD constant at each level

Case 3: WD decreases down the tree

Q1: $T(n) = 9T(n/3) + n$ $\rightarrow$ cost at single step

$a=9, b=3, f(n)=n$

* make watershed func. i.e. $n^{\log_b(a)}$

Plug the values in watershed:

$$n^{\log_3(9)} = \boxed{(n)^2}$$

* As watershed $(n^2) > f(n)$ i.e. $n \therefore$ it's case 1.

* n^2 should provide upper bound to $f(n)$ i.e. ' n ' with specific epsilon.

$$f(n) = O(n^{\log_b(a) - \epsilon}) = O(n^{2 - \epsilon})$$

* Finding ϵ : Epsilon should always be ~~less~~ ^{lesser} than or equal to 1 here as ~~above~~ ^{above} 1 will cause $\rightarrow n^{2-1.5} = (n)^{0.5}$.
 This fails the condition as ~~(n)~~ ⁽ⁿ⁾ $(n)^{0.5}$ does not provides upper bound to ' n '. $\therefore$

$$T(n) = O(n^2) \text{ with } \epsilon \leq 1.$$

Q4: $3T(n/4) + n \log n$. $a=3$, $b=4$, $f(n) = n \log n$
 watershed $\rightarrow n^{\log_4(3)} = n^{0.8}$

* Since power of watershed is 0.8 & highest power of $f(n)$ is '1' which is of n (as $n^1 \log n$) $\therefore$ we'll provide Case 3 as watershed $< f(n)$

$$f(n) = \Omega(n^{0.8 + \epsilon}) \quad \therefore \epsilon \leq 0.2 \quad \left(\text{for it to be less than } n^1 \text{ (ignore } \log n) \right)$$

~~Reg.~~ Reg. Constraint: $T(n) = O(n \log n)$
 $3 \left(\frac{n}{4} \log \frac{n}{4} \right) \leq c \cdot n \log n$

$$\boxed{c < 1}$$

NOTE: if $f(n) = n \log n$ & watershed = n ; we'll have case #2 as max powers of 'n' are same i.e. 1. This is because there is no "polynomial times growth"; we're logarithmic times growth (not imp.)

$\Rightarrow$ Substitution:

Q: $T(n) = 2T(n/2) + n \rightarrow (1)$. $T(1) = 1$

$T(n/2) = 2T(n/4) + n/2$

* Put value of $T(n/2)$ in (1)

$$T(n) = 2 \left[2T\left(\frac{n}{4}\right) + \frac{n}{2} \right] + \frac{n}{2}$$

$$T(n) = 4T(n/4) + 2n \rightarrow (2)$$

$$T(n/4) = 2T(n/8) + (n/4)$$

* Put $T(n/4)$ in (2)

$$T(n) = 4\left(\frac{n/2}{2} + \frac{n}{4}\right) + 2n$$

$$T(n) = 8T\left(\frac{n}{8}\right) + 3n \longrightarrow 8T\left(\frac{n}{2^3}\right) + 3n$$

* At 'k' Step:

$$T(n) = (2^k)T\left(\frac{n}{2^k}\right) + kn \longrightarrow (A)$$

* At Base case :- $\boxed{n/2^k = 1}$ (as Base case $\rightarrow T(1) = 1$)
 $n = 2^k$
 $k = \log_2(n)$

* put k in A so time complexity = $kn \rightarrow \boxed{n \log n}$